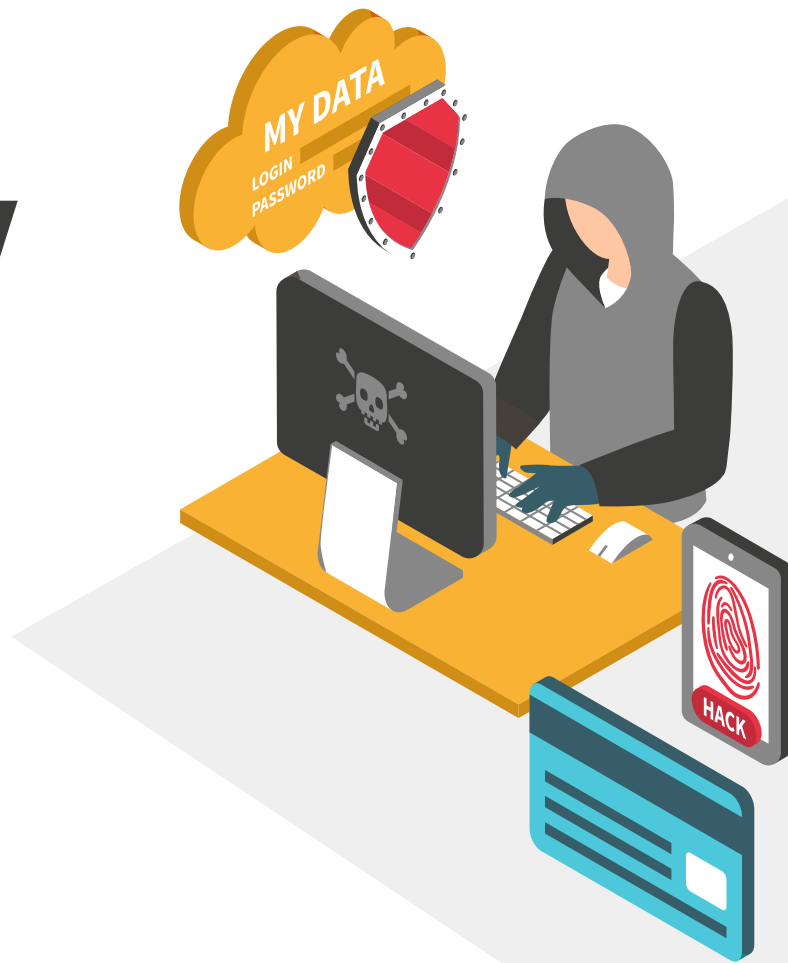




CYBERSECURITY **THREATS** ARE INCREASING RAPIDLY

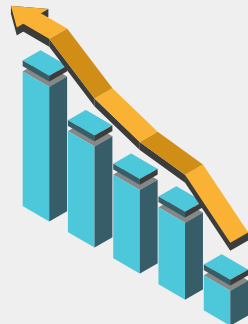
Lam Yoke Yu A23CS0233

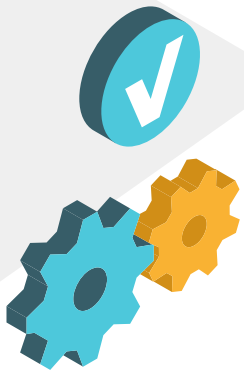




19.62M

Web-based attacks in the first half of the year of 2024





**TOO REAL
TO BE FAKE**





01.

BACKGROUND OF THE PROBLEM

History of cybersecurity



BACKGROUND OF THE PROBLEM

PHONE PHREAKING

“Phone phreaking” emerged

1950



1960s

IBM CTSS

The IBM CTSS system revealed one of the first known software vulnerabilities

CREEPER

Creeper, the first computer virus, spread via ARPANET

1971



BACKGROUND OF THE PROBLEM

MACRO VIRUSES

Macro viruses surged, prompting security tools like SSL



1986

1990s

2000s

MARCUS HESS

Marcus Hess infiltrated over 400 military and research computers



INCREASE SOPHISTICATION

Malware, phishing, ransomware, and social engineering attacks have escalated in scale and sophistication



02.

CAUSE OF THE PROBLEM

Rationalisation, financial motivation, system vulnerabilities



CAUSE OF THE PROBLEM



RATIONALISATION

Rationalise their actions using values praised in modern society



FINANCIAL MOTIVATION

Cybercriminals exploit system vulnerabilities and human gullibility to obtain monetary benefits



SYSTEM VULNERABILITIES

Software vulnerabilities, network protocol shortcomings, device proliferation



03.

EFFECTS OF THE PROBLEM

Financial loss, psychological impact, reputational damage



EFFECTS OF THE PROBLEM



FINANCIAL LOSS

In the first quarter of 2024, police recorded 12,110 online fraud cases in Malaysia, involving losses of RM573.7 million



PSYCHOLOGICAL EFFECT

Victims may suffer from trauma, stress, anxiety or distrust of digital systems



REPUTATIONAL DAMAGE

Facebook data breach in 2019 and 2021



04.

POSSIBLE SOLUTIONS

Education and awareness, integrating cybersecurity into system design



EDUCATION & AWARENESS

IMPLEMENTATION

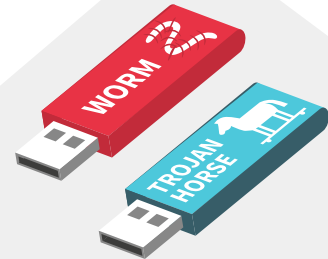
- Raise public awareness about digital risks
- Train individuals to recognize and avoid scams
- Teach students to importance of ethical behaviour

ADVANTAGES

- Promote a culture of cybersecurity awareness
- Reduce vulnerability
- Encourages responsible behaviour

DISADVANTAGES

- Not all individuals retain or apply the knowledge in real-life situation
- Changing psychological and social influences



INTEGRATING CYBERSECURITY INTO SYSTEM DESIGN

IMPLEMENTATION

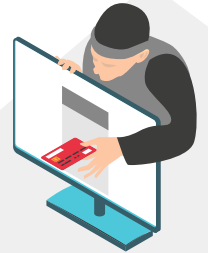
- Train developers in secure coding practices
- Make knowledge in cybersecurity mandatory for all developers

ADVANTAGES

- Stronger and more secure systems for the start
- Reduce the number of exploitable vulnerabilities

DISADVANTAGES

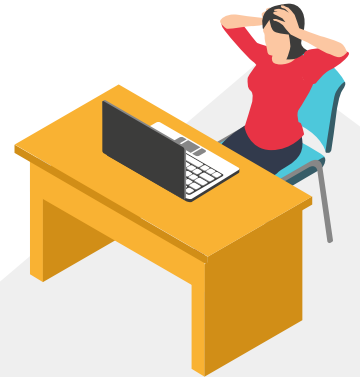
- Difficult to keep up with the rapid technological advancement
- Require more time and cost and require expertise





05.

CONCLUSION





THANKS

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